Week 9 — Gas-Powered Carts & Special Systems

Sessions: 1 (Theory) · 2 (Lab) · **Total:** 4 hours
Primary LOs: LO-11
Hands-on target: Session 2 ≈ 75% practical (+ strong safety brief)

Instructor Preparation Guide (Complete Day Before)

Pre-session setup

- [ ] Stage 1–2 gas carts or engine stand (Subaru/Kohler/typical single-cylinder 4-stroke)
- [ ] Fire extinguisher at station; no smoking zone; fuel handling SOP posted
- [ ] Compression tester, spark tester (inline/safe type), plug socket, feeler gauge
- [ ] Fresh spark plugs for demo; sample fouled plugs
- [ ] Print `labs/W09_gas_diagnostic_checklist.md`
- [ ] Verify kill switch / seat switch operation before student work
- [ ] If EFI cart available: demo only unless instructor EFI-qualified

Planted / selected conditions

Condition	Notes
Fouled plug	Classic no-start teaching aid
Restricted fuel filter	If safely recreatable
Low compression engine (known)	For comparison readings
Misadjusted governor (instructor demo only)	Do not let students set unsafe high RPM

Safety staging

- Cool engines before plug removal when possible; hot exhaust warning.
- Carbon monoxide: doors/ventilation; never run indoors without exhaust plan.
- Fuel spills: stop, absorb, ventilate — no starting.

Week 9 Learning Objectives

1. Explain basic 4-stroke operation in gas golf carts.
2. Perform basic diagnostics: compression, spark, fuel delivery checks.
3. Recognize when issues need engine-specialty expertise vs electrical-style isolation.

Session 1 — Theory & Demonstration (2 hours)

Materials needed: Gas cart or engine stand; sample plugs (good/fouled); fuel filter cutaway if available; extinguisher at instructor station; 4-stroke diagram; ventilation confirmation.

Timed Agenda

Time	Min	Activity	Instructor Notes
0:00	10	Gasoline safety deep-dive	Fire, fumes, hot surfaces, spill response
0:10	15	4-stroke basics: intake/compression/power/exhaust	**[DIAGRAM PLACEHOLDER: 4-stroke cycle for single-cylinder cart engine]**
0:25	15	Fuel system: tank, filter, pump, carb/EFI, governor	
0:40	15	Ignition: plug, coil, kill/safety interlocks	Seat/brake switches common
0:55	5	Break	
1:00	20	Symptom map: no-start, surge, power loss, overheat, smoke	
1:20	15	Electric vs gas diagnostic mindsets compared	Different tools; same 7 steps
1:35	15	Fleet trend note: gas → electric/lithium conversions	Maintenance cost/emissions context
1:50	10	Lab PPE & extinguisher locations check	

Key Teaching Points

- Same 7-step process — different subsystem tests (compression/spark/fuel).
- Safety interlocks on gas carts strand “good engines.”
- Governor mistakes can overspeed and destroy engines — instructor-controlled adjustments only.
- EFI requires different tooling; know your limits.

Common Misconceptions

Misconception	Correction
“No-start = bad engine.”	Kill switch, empty tank, fouled plug, safety switch often guilty.
“More fuel = better.”	Flooding and mixture issues cause misfire/smoke.
“Gas cart electrical doesn’t matter.”	Charging/lighting/solenoids/safety circuits still critical.

Safety Reminders

- Extinguisher within reach before cranking after fuel work.
- No open containers of gasoline in bay.
- Disconnect spark lead properly when pulling compression if procedure requires.

Session 2 — Hands-On Lab (2 hours)

Materials needed: `labs/W09_gas_diagnostic_checklist.md`; compression tester; safe spark tester; plug tools; rags/absorbent; extinguisher; PPE; instructor EFI demo only if equipped.

Timed Agenda

Time	Min	Activity	Instructor Notes
0:00	10	Safety brief + extinguisher + ventilation check	Sign safety acknowledgment on checklist
0:10	20	Compression test (training cart/stand)	Record psi; discuss dry/wet if taught
0:30	20	Spark test (safe method) + plug inspection/gap	Compare fouled vs new
0:50	5	Break	
0:55	20	Fuel system basics: filter condition, fuel present, bowl check if accessible	No smoking; rags ready
1:15	15	Safety interlock quick checks (seat/brake/kill)	
1:30	15	Side-by-side: write how 7-step differed from electric cart	
1:45	15	Cleanup; quiz; secure fuel caps	

Assessment Focus

Gas diagnostic checklist + short quiz on engine fundamentals vs electric systems.

Week 9 Quiz (8 questions)

1. A typical golf cart gas engine in this curriculum is described as:
   A) V8 diesel
   B) Single-cylinder 4-stroke (common OEM families)
   C) Turbine
   D) Two-stroke outboard only

2. Before diagnosing a no-start gas cart, verify:
   A) Only tire brand

B) Fuel present, spark potential, compression path, and safety interlocks
C) Lithium cell balance
D) PWM percent

3. Running a gas cart engine indoors without exhaust control risks:
   A) Better fuel economy
   B) Carbon monoxide exposure
   C) Higher SG
   D) Stronger regen

4. Governor systems primarily:
   A) Control maximum engine speed / load response
   B) Charge lithium packs
   C) Replace brakes
   D) Read Curtis codes

5. True or False: The 7-step diagnostic process still applies to gas carts.

Short Answer

1. List the three classic no-start pillars checked in lab (spark/fuel/compression concept).
2. Name two gas-cart safety interlocks commonly found.
3. Why might a fleet convert gas carts to electric/lithium?

Answer Key

1. **B**
2. **B**
3. **B**
4. **A**
5. **True**
6. Spark, fuel delivery, compression (plus air & interlocks as taught)
7. Seat switch, kill switch, brake/pedal switch, neutral start — any two valid
8. Lower operating cost, less emissions/maintenance, shop standardization, customer preference, etc.